2026

North-West University

Engineering Design 2026 course team

[SYSTEM SPECIFICATION]

Hoisting System — Engineering Design 2026

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Origination and Approval

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Acceptance

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Distribution List

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1 Introduction and scope

1.1 Identification

The system is a rail-mounted gantry/trolley hoisting platform, reusing the existing I-beam rail and gantry structure. A trolley moves laterally along the rail (MCTR I) and hoists a payload via a spooling mechanism (MCTR II), driven by an induction motor powered through custom drive electronics (EEII I). The payload is engaged via a custom electromagnetic attachment (EEII II). All subsystems interface with a shared embedded controller (ESP32-S3-WROOM-1 core board plus an isolated IO board), which executes real-time control locally and links wirelessly (nRF24L01) to a PC-based dashboard (REII) for monitoring, control, and motion-profile shaping.

The system decomposes into five work packages coordinated by this shared embedded controller, supplied to every group regardless of computer-engineering staffing: MCTR I (Lateral Movement), MCTR II (Spooling), EEII I (Drive Electronics), EEII II (Magnetic Load Attachment), and REII (Remote Control Dashboard).

1.2 Intended use

This document is an initial, high-level requirements specification for the 2026 Engineering Design hoisting system (Rev 0 — for discussion; bracketed [TBD] values are placeholders pending detailed design). It sits at the operational-to-technical level: it defines what the system and its work packages must do, not how each work package implements its solution — detailed component-level requirements are left to each work package's own design documentation (the Controller Specification, and the five work-package Scopes of Work listed in Section 2.1).

The system shall be operable via the PC-based dashboard where a group has computer-engineering capacity, and via the controller's standalone Wi-Fi/UDP interface where it does not (see 4.2.2 Modes).

1.3 Intended users

The system is intended to be built and operated by the Engineering Design 2026 student groups responsible for each work package, and demonstrated to course assessment staff at the end of the semester. Operators interact with the system either through the REII dashboard or, for groups without computer-engineering capacity, directly through the controller's standalone interface.

2 Applicable and other referenced documents

2.1 Applicable documents

Table 1: Applicable documents

Document Identifier	Document Description	Document Location
Controller Specification	Defines the shared embedded controller (ESP32-S3-WROOM-1 core board, IO board, isolation strategy, pin assignment) referenced throughout this specification.	Engineering Design 2026 project folder
MCTR I – Lateral Movement (SOW)	Work-package scope of work for the lateral movement mechanism.	Engineering Design 2026 project folder
MCTR II – Spooling (SOW)	Work-package scope of work for the hoist/spooling mechanism.	Engineering Design 2026 project folder
EEII I – Drive Electronics (SOW)	Work-package scope of work for the motor drive electronics.	Engineering Design 2026 project folder
EEII II – Magnetic Load Attachment (SOW)	Work-package scope of work for the electromagnetic load attachment.	Engineering Design 2026 project folder
REII – Remote Control Dashboard (SOW)	Work-package scope of work for the RF ground station and dashboard software.	Engineering Design 2026 project folder

2.2 Other referenced documents

Table 2: Other referenced documents

Document Identifier	Document Description	Document Location
Espressif ESP32-S3-WROOM-1 Datasheet	Governs GPIO count, strapping pins, and native-USB routing assumed by DR5 and the controller pin budget.	Espressif Systems (public)
Texas Instruments ADS1220 Datasheet	Governs the 2.048V internal reference used to bound single-ended current-sense signals in IR1/IR2.	Texas Instruments (public)
"2.2 EngDes26 – Requirements and Specs" (course slide deck)	Source of the requirements-engineering methodology (functional/performance/MOE/MOP/MOS taxonomy) applied throughout this document.	NWU Engineering Design 2026 course materials

3 Definitions, Acronyms, and Abbreviations

3.1 Definitions

As specified means as specified in Section 4 of this document.

May expresses permissive guidance.

Shall expresses a binding requirement.

State expresses an observable condition of the system at a given time.

Mode expresses behaviour exhibited by the system.

Work package means one of the five discipline-scoped deliverables (MCTR I, MCTR II, EEII I, EEII II, REII) into which the system's design and build effort is divided.

The Oxford Concise English Dictionary shall be used in the interpretation of terms in Section 4, Requirements, which are not otherwise defined above.

3.2 Acronyms

Acronym	Meaning
MCTR	Mechatronics (work-package prefix)
EEII	Electrical and Electronic Engineering II (work-package prefix)
REII	Remote-control Engineering / dashboard work package
FR	Functional Requirement
NFR	Non-Functional Requirement (superseded in this document by Other Qualities / Physical requirements per template structure)
DR	Design Requirement
IR	Interface Requirement
PT	Performance Target
MOP	Measure of Performance
MOS	Measure of Suitability
ADC	Analog-to-Digital Converter
PWM	Pulse-Width Modulation
RF	Radio Frequency
RCD	Residual-Current Device
COTS	Commercial Off-The-Shelf
UDP	User Datagram Protocol
HV	High Voltage

Acronym	Meaning
AC / DC	Alternating Current / Direct Current
NWU	North-West University
TBD	To Be Determined

3.3 Abbreviations

None beyond the acronyms listed in 3.2.

4 Requirements

4.1 Identification of external interfaces

Interface	Description
Controller ↔ EEII I (Drive Electronics)	Isolated 6-line gate/PWM interface plus one reserve line; single-ended current-sense input. See IR1.
Controller ↔ EEII II (Magnetic Load Attachment)	Isolated PWM interface and single-ended current-sense input (≤2.048V). See IR2.
Controller ↔ MCTR I (Lateral Movement)	Direction, PWM, and quadrature-encoder interface. See IR3.
Controller ↔ MCTR II (Spooling)	Load-cell input and brake-control interface. See IR4.
Controller ↔ REII (Dashboard)	Wireless nRF24L01 link carrying telemetry and control commands; comms protocol is REII's own design.
Controller ↔ Expansion header	5-pin, protocol-agnostic, low-voltage digital logic header for group-specific extensions. See IR5.

4.2 Identification of states and modes

4.2.1 States

- Idle — system powered but not executing a commanded motion, hoist, or magnet-engagement action.
- Operational — system actively executing a commanded lateral, hoist, or magnet-engagement function under either mode below.
- Fault / E-Stopped — the hardware e-stop has removed power from the HV bus (see PHY2); all motive functions are inhibited independent of software or firmware state.

4.2.2 Modes

- Dashboard-controlled — the operator controls the system via the REII PC-based dashboard over the nRF24L01 link.
- Standalone — for groups without computer-engineering capacity, the operator controls the system directly via the controller's built-in Wi-Fi/UDP interface, without a dashboard.

4.3 System function and performance requirements

Requirements are grouped by owning work package, and split across tables so each stays readable on one page. Each requirement is a single, testable statement per the parsing-template discipline (actor, condition, action, object, constraint). Where a requirement has a quantitative performance dimension, it is deliberately written as a functional pass/fail threshold only — the associated performance target is separated out below. This reflects experience that functional correctness, not raw performance, is the harder and more common failure point: a system that merely functions should pass, with performance differentiating marks rather than gating them.

4.3.1 Mechanical (MCTR I, MCTR II)

ID	Requirement
FR1	The lateral movement mechanism, upon receipt of a motion command from the controller, shall move the trolley along the I-beam rail in the commanded direction, using a COTS DC motor costing less than R250 (MCTR I). See PT1 for the associated performance target.
FR2	The lateral movement mechanism shall report wheel position to the controller via quadrature encoder feedback, to support motion profiling by REII (MCTR I).
FR3	The lateral movement wheels shall be custom-manufactured by 3D printing, supporting the wheel wear-characterization exercise (MCTR I).
FR4	The spooling mechanism, upon receipt of a raise or lower command, shall drive the hoist gearbox to lift or lower a payload of up to 20kg in the commanded direction (MCTR II). See PT2 for the associated performance target.
FR5	The spooling mechanism shall report on-hook load to the controller continuously during operation (MCTR II).
FR6	The active brake shall engage automatically upon loss of drive power or command signal, holding the payload without drift (MCTR II).

4.3.2 Electrical (EEII I, EEII II)

ID	Requirement
FR7	The drive electronics, upon receipt of gate/PWM commands from the controller, shall convert single-phase 230VAC mains to a DC bus and invert this to a 3-phase drive for the induction motor (EEII I).
FR8	The drive electronics shall detect and interrupt drive output upon shoot-through or overcurrent conditions, protecting the switching devices from damage (EEII I).
FR9	The drive electronics shall provide a single-ended current-sense signal, capped below 2.048V, to the controller for firmware-based fault inference (EEII I).
FR10	The magnetic load attachment, upon receipt of an engage command, shall generate sufficient holding force via a 3-point electromagnet to retain a payload of up to 20kg (per DR6) under static conditions (EEII II). See PT3 for the associated performance target.
FR11	The magnetic load attachment shall report an on-load indication to the controller, derived from current-sense discrimination between a loaded and unloaded grip (EEII II).

4.3.3 Software and Controller (REII, cross-cutting)

ID	Requirement
FR12	The dashboard, upon receipt of telemetry from the controller via the nRF24L01 link, shall display all operational parameters (position, speed, load, current, fault, grip status) to the operator (REII).
FR13	The dashboard shall provide the operator full control of all system operations, including motion, hoist, and magnet-engagement commands (REII).
FR14	The embedded controller shall execute its real-time control loop — drive signal generation, sensor acquisition, safety interlocks — independently of wireless link or dashboard availability.
FR15	For groups without computer-engineering capacity, the controller shall accept motion and control commands via UDP over its built-in Wi-Fi, requiring no additional hardware.

4.3.4 Performance targets

Graduated marking criteria, not pass/fail gates — each links back to a functional requirement above. Marks scale with how closely the target is achieved rather than requiring it outright. Consistent with treating these as Measures of Performance (directly quantifiable, expressible on a scale) rather than binary compliance checks.

ID	Linked FR	Target	Marking Basis
PT1	FR1 (Lateral movement)	≥ 1 m/s, measured at 1m from the departure point, unloaded	Full marks at target; marks scale down proportionally for lower demonstrated speed.
PT2	FR4 (Spooling)	1.6m hoist stroke in ≤ 20s (≈ 80mm/s), with a 20kg payload	Full marks at target; marks scale down proportionally for a slower demonstrated rate.
PT3	FR10 (Magnetic attachment)	Retain a 20kg payload against a disturbance force [TBD]	Full marks at target; marks scale down proportionally below threshold. Numeric threshold pending further dynamics modelling.

4.4 Relationship between states and modes

Both operating modes (Dashboard-controlled and Standalone) are available whenever the system is in the Operational state; mode selection does not affect what states the system can enter. The Fault/E-Stopped state pre-empts both modes, inhibiting motive output regardless of which mode was active when the e-stop engaged. The system begins in the Idle state at power-up, before an operator issues a first command under either mode.

4.5 System external interface requirements

Listed in the same order as the interfaces identified in 4.1.

ID	Requirement
IR1	The controller shall present an isolated 6-line gate/PWM interface to the drive electronics for 3-phase motor control, plus one reserve line.
IR2	The controller shall present an isolated PWM interface and a single-ended current-sense input (≤2.048V) to the magnetic load attachment.
IR3	The controller shall present a direction, PWM, and quadrature-encoder interface to the lateral movement mechanism.
IR4	The controller shall present a load-cell input and a brake-control interface to the spooling mechanism (channel to be finalized — ADS1220 differential input or dedicated HX711).
IR5	The controller shall present a 5-pin, protocol-agnostic expansion header for group-specific extensions, documented as low-voltage digital logic only.

Full pin-level detail (module pins, edge-header assignment, isolation strategy) is maintained in the Controller Specification document, not duplicated here.

4.6 Environmental requirements

Deployment: the system shall operate on the existing I-beam rail and gantry structure, located outdoors on the North-West University North Campus.

The system shall operate outdoors, exposed to ambient weather conditions typical of the NWU North Campus — temperature variation, precipitation, solar/UV exposure, wind, and dust — and in the presence of other groups' concurrently operating systems on shared or adjacent rail infrastructure.

Electrical enclosures and wiring shall provide ingress protection against rain, dust, and incidental moisture, given the system's outdoor deployment; exposed materials, including any 3D-printed or polymer components, shall be selected or rated for sustained outdoor UV and temperature exposure across the semester.

Operational life-cycle: the system shall remain operational through the design, build, integration, and demonstration phases of the 2026 academic semester.

4.7 External resource requirements

Maintains the same order as 4.1 / 4.5 — resources enter or exit the system only via the interfaces already identified.

The system shall receive electrical power via a single dedicated 1-phase, 50Hz, 230VAC mains supply point per group. Distribution of this supply to all subsystem voltage rails (e.g. 3V/5V/12V/24V, each independently selected COTS supplies by their owning work package) and provision of the main on/off switch is a system-level integration responsibility, assigned to EEII I.

4.8 Physical requirements

ID	Requirement
PHY1	The payload used to test and demonstrate the system shall present a flat sheet-metal attachment surface of 600mm x 400mm, at least 3mm thick, for engagement by the magnetic load attachment. This surface may be supplemented with additional mass to raise the total payload to a maximum of 20kg, provided the combined centre of mass remains centred on the 600mm x 400mm attachment surface. (Formerly DR6; the general system mounting plate is unaffected by this requirement and is not required to be sheet metal.)
PHY2 (Safety)	The system shall include a hardware e-stop wired directly into the HV bus, capable of removing power from the drive electronics independent of software or firmware state.
PHY3 (Safety)	The mains supply feeding the drive electronics shall be protected by a residual-current device (RCD), given the combination of HV wiring and outdoor exposure to moisture.

4.9 Other qualities

ID	Requirement
OQ1 (Reliability)	Shoot-through and overcurrent protection shall be validated on a milled test board prior to full-scale VFD assembly, directly addressing the prior year's dominant failure mode.
OQ2 (Maintainability)	The core controller board shall be designed as reusable general-purpose lab infrastructure, providing local analog acquisition and data logging independent of any single year's design problem.
OQ3 (Usability)	The dashboard interface shall present telemetry and controls such that an operator can safely command the system after a brief orientation, with no specialist RF or embedded-systems knowledge required.
OQ4 (Affordability)	Component costs shall not exceed the stated constraints for each work package (e.g. lateral-movement motor < R250); total system cost per group shall not exceed R5,000, inclusive of all work-package procurement.

4.10 Design and construction requirements

Design requirements constrain the solution space ahead of detailed design. Per requirements-engineering practice, each is included only with an explicit rationale, since design requirements should be used conservatively.

ID	Requirement	Rationale
DR1	The lateral movement drive motor shall be a COTS DC motor costing less than R250.	Bounds procurement cost and lead time across all groups; keeps motor selection a sizing exercise rather than an open-ended search.
DR2	The spooling gearbox shall not use a worm gear configuration.	Forces a multi-stage gearbox design exercise consistent with the programme's mechanical design intent, rather than a single high-ratio COTS-equivalent solution.
DR3	The magnetic load attachment shall not use a complete COTS electromagnet or solenoid; COTS core material is permitted.	Preserves hands-on electromagnetic design and manufacturing experience while keeping raw-material procurement low-risk.
DR4	The wireless link shall use the nRF24L01 module.	Standardizes the wireless link across every group and the provided controller firmware, so the fixed low-level link behaviour is consistent everywhere.
DR5	The embedded controller shall be based on the ESP32-S3-WROOM-1 (N8R2) module.	Standardizes the platform provided to every group regardless of computer-engineering staffing, ensuring consistent GPIO and ADC capability.

5 Notes

5.1 Verification requirements

Each requirement above will be assigned an explicit verification method (how compliance is measured, and with what) in the next revision of this document, following the Measure of Performance / Measure of Suitability framework: the Section 4.3.4 performance targets (PT1–PT3) are Measures of Performance; supportability-type requirements (e.g. OQ1's protection validation, OQ2's maintainability) are Measures of Suitability. Functional requirements in Section 4.3 are verified as pass/fail — demonstrated or not — independent of the graduated scoring applied to their linked performance targets. Verification requirements will be written with the same rigor as the requirements themselves — explicit, measurable, and tied to a defined test method or standard — rather than left implicit.

5.2 List of safety requirements

Duplicated from Section 4.8 (Physical requirements) for visibility; see 4.8 for the authoritative statement.

- PHY2 (Safety) — The system shall include a hardware e-stop wired directly into the HV bus, capable of removing power from the drive electronics independent of software or firmware state. (See 4.8.)
- PHY3 (Safety) — The mains supply feeding the drive electronics shall be protected by a residual-current device (RCD), given the combination of HV wiring and outdoor exposure to moisture. (See 4.8.)

5.3 Open items

The disturbance-force threshold in PT3 remains open pending further dynamics modelling. Payload mass is fixed at up to 20kg per PHY1, and the lateral/hoist performance targets are fixed per PT1/PT2. The spooling load-cell's ADC path (ADS1220 differential channel vs. dedicated HX711) remains an open implementation choice per the Controller Specification and is not treated as a hard requirement here.

Document History

Issue	Date	Status	Filename
0	2026-08-04	For discussion	System_Requirements_Specification_2026

Revision History

Issue	Date	Changes
0	2026-08-04	Initial high-level draft, per the Engineering Design 2026 requirements-engineering methodology.

Authorisation History

Issue	Date	Status	Reference

EEII I
Drive Electronics

Work Package Scope of Work — Engineering Design 2026

> **CONSTRAINT: Protection against shoot-through and overcurrent is mandatory**

Main Work Elements

1. **Full VFD chain design —** a simple rectifier and bulk capacitors for the HV DC bus, plus single-phase-to-3-phase inversion (discrete MOSFET/IGBT with gate drivers) driving the induction motor. Architecture is largely carried over from 2025's approach.
2. **Component selection —** switching devices, gate driver ICs, and a current-sense resistor feeding the shared ADC channel.
3. **Protection circuit design and validation —** the Increment 2 half-bridge PCB milling exercise — students mill a board from a supplied component set and use it to validate shoot-through and overcurrent protection before committing to the full 3-phase board.
4. **Simulation —** switching behaviour, protection response, and thermal headroom ahead of hardware commitment.
5. **E-stop integration —** a physical e-stop wired directly into the HV bus so activation reliably kills the system's main dangerous component.
6. **System power integration —** the main on/off switch and mains distribution to the HV bus and to each subsystem's already-selected COTS low-voltage supply. Integration and switching only — each work package selects its own supply.
7. **Controller interface —** the six isolated gate/PWM lines plus reserve line; current-sense conditioned to a single-ended signal capped below 2.048V (the shared ADC's internal reference), occupying one ADC channel; fault inferred in firmware from commanded-vs-measured mismatch rather than a dedicated hardware flag.
8. **Full build and test —** assembling the complete VFD board, verifying it drives the induction motor, confirming protection trips under fault injection, and verifying e-stop and power distribution.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).

EEII II
Magnetic Load Attachment

Work Package Scope of Work — Engineering Design 2026

> **CONSTRAINT: No complete COTS electromagnets (COTS core material is permitted)**

Main Work Elements

1. **Electromagnet design and simulation —** magnetic circuit design for the 3-point attachment — core geometry, air gap, turns and current — sized to meet the required holding force for the payload.
2. **Magnet manufacturing —** hand-wound coils for all 3 attachment points; COTS core material is permitted, but no pre-made electromagnets or solenoids may be procured.
3. **Drive electronics —** a switching circuit to energize and de-energize the magnet, including selection of its own COTS low-voltage supply.
4. **On-load indicator via current-sense —** characterizing the current signature difference between gripping a load and open air, and calibrating detection thresholds.
5. **Controller interface —** the isolated magnet PWM line and the current-sense signal, single-ended and capped below 2.048V, into the header already defined.
6. **Testing and validation —** confirming holding force under real load and validating the on-load indicator's reliability across repeated grip/release cycles.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).

MCTR I
Lateral Movement

Work Package Scope of Work — Engineering Design 2026

> **CONSTRAINT: DC motor cost < R250**

Main Work Elements

1. **Mechanism design —** trolley/carriage on the rail, wheel and bearing mounting, custom 3D-printed wheels (feeding the wheel wear-characterization exercise), other motion components (gears, pulleys, belts) printed or bought as practical, and the overall system mounting plate as a direct output of this design work.
2. **Motor and driver selection —** COTS DC motor under the R250 constraint with a matching driver, including selection of its own COTS low-voltage supply.
3. **Drivetrain dynamics and sizing —** acceleration/deceleration modelling to justify the motor and driver choice, including the centre-of-mass and weight-distribution analysis that also drives the mounting plate layout.
4. **Controller interface —** wiring direction, PWM, and quadrature encoder signals into the IO board header already defined; confirming encoder resolution is adequate for the dashboard's motion profiles.
5. **Characterization and test —** measuring actual acceleration/deceleration against the modelled prediction, confirming cost-constraint compliance, correlating wheel wear against print parameters, and handing off the validated performance envelope to REII.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).

MCTR II
Spooling

Work Package Scope of Work — Engineering Design 2026

> **CONSTRAINT: No worm gearbox**

Main Work Elements

1. **Hoist gearbox spec and dev —** minimum 2-stage gearbox for the spool, driven off EE1's induction motor, no worm gearbox permitted.
2. **Active brake design —** fail-safe holding brake satisfying the inherently-safe requirement, its control interface back to the controller, and selection of its own COTS low-voltage supply rail.
3. **On-hook load measurement —** load cell in the hoist path, feeding telemetry and interlock use (via an ADS1220 differential channel or a dedicated HX711 on the spare IOs — not yet committed).
4. **Dynamics simulation —** lift/lower dynamics — load, speed, gear ratio, and brake engagement behaviour — validating the mechanism against required performance.
5. **Controller and safety interface —** wiring load cell and brake signals into the controller, and defining interlock logic tying brake engagement to load and fault conditions.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).

REII
Remote Control Dashboard

Work Package Scope of Work — Engineering Design 2026

> **CONSTRAINT: Must use the nRF24L01 module**

Main Work Elements

1. **RF link bring-up —** building the STM32F411 ground-station dongle from scratch — native USB, nRF24 driver, USB bridging, with no scaffolding provided — and completing the stubbed nRF24 SPI driver on the ESP32 controller side. Ground-station hardware is USB-bus-powered from the PC, drawing nothing from the shared system power.
2. **Comms protocol design —** defining the application-layer protocol over the raw nRF24 link — framing, addressing, and any reliability mechanism such as ACK/retry — as open design space rather than a fixed specification.
3. **Dashboard software —** live monitoring of all operational telemetry and full command and control of system operations.
4. **Anti-sway motion-profile modelling —** modelling the payload/pendulum dynamics and deriving ramp-rate or velocity-shaping parameters, sent to the controller as open-loop feedforward profiles executed on MCTR I's drive.
5. **Obstacle-avoidance path planning —** the same open-loop, planned-navigation pattern, generating motion commands around known obstacles.
6. **Testing and validation —** validating dashboard behaviour against actual crane operation and confirming the anti-sway and obstacle-avoidance profiles perform as modelled.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).